Miniproject 2: 4-bit shift register

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https://github.com/James-Jagielski/MADVLSI/tree/main/Miniproject2

1 Schematics

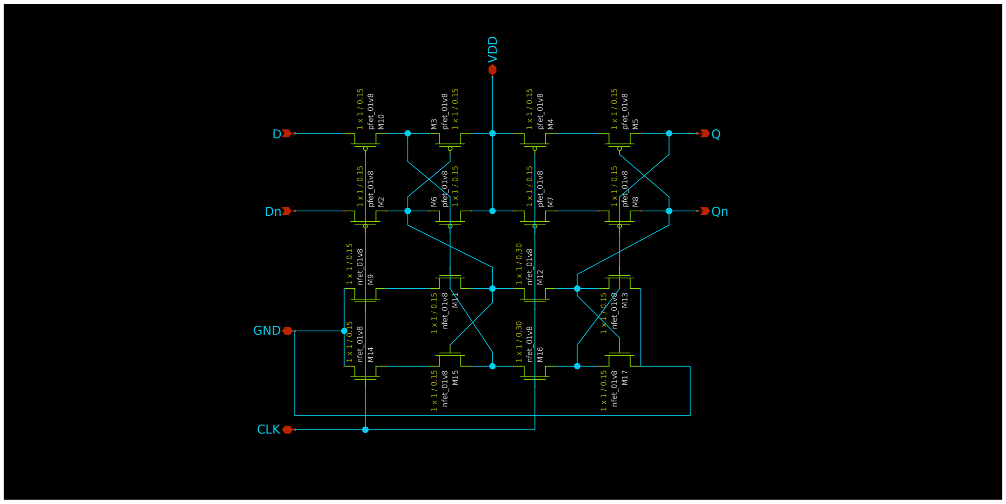

Figure 1: CSRL latch schematic.

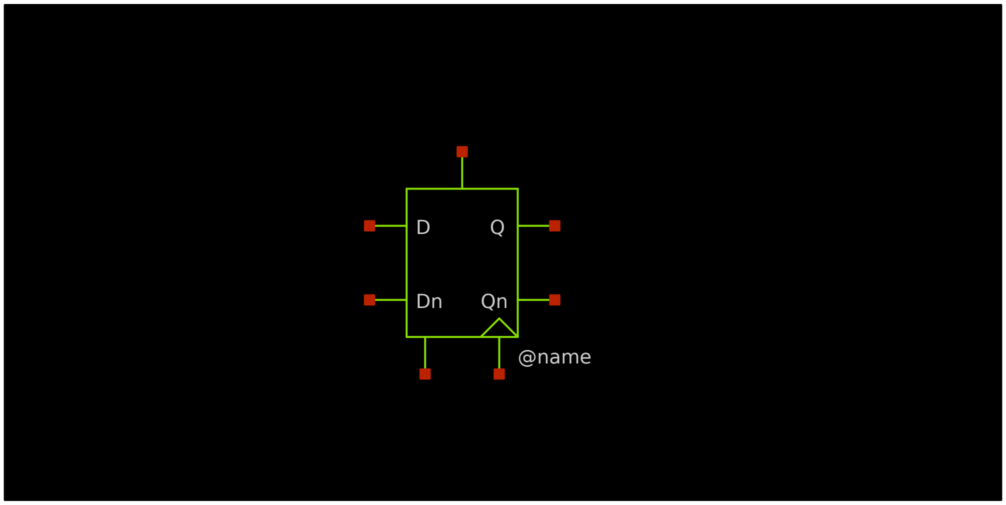

Figure 2: Latch symbol.

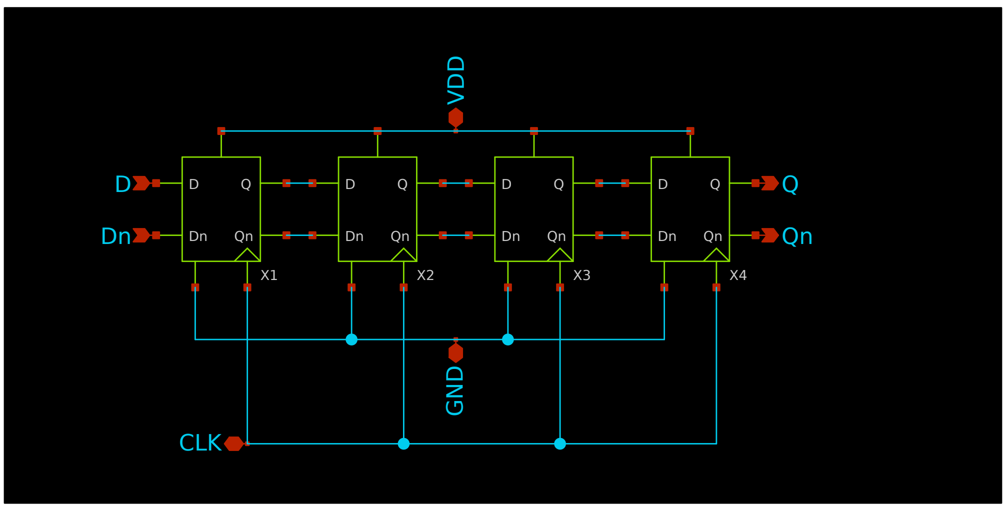

Figure 3: 4 bit shift register schematic.

2 Simulations

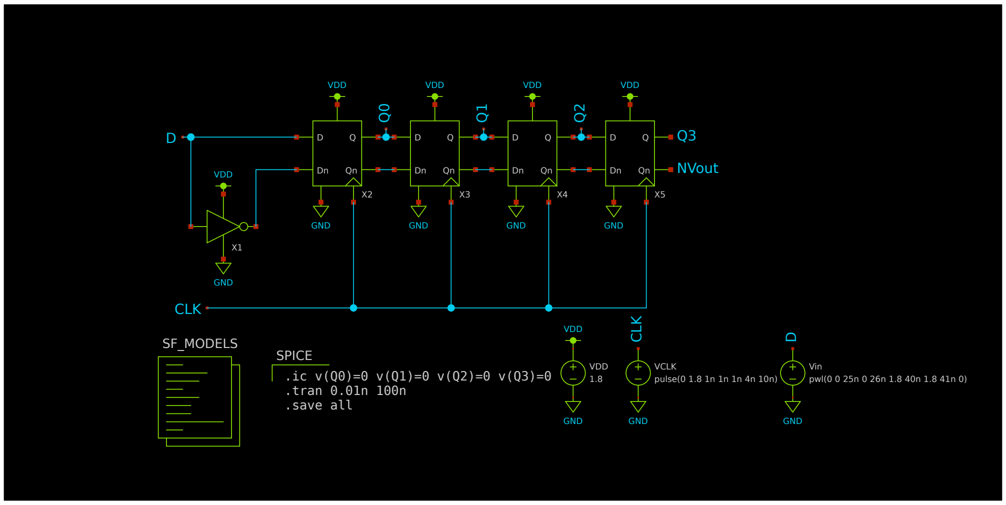

Figure 4: Shift register simulation schematic.

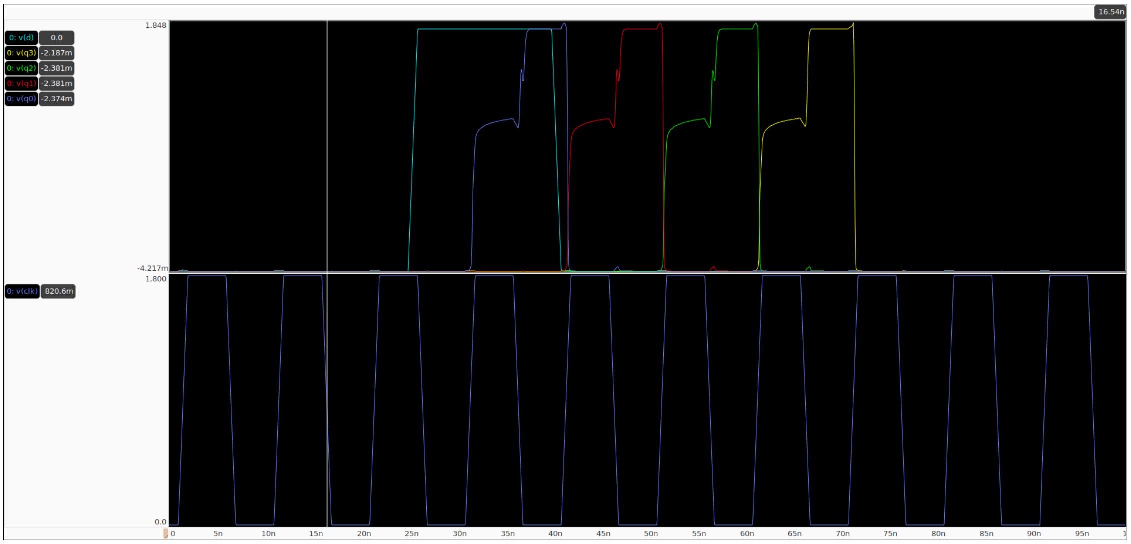

Figure 5: Shift register simulation results.

Layout

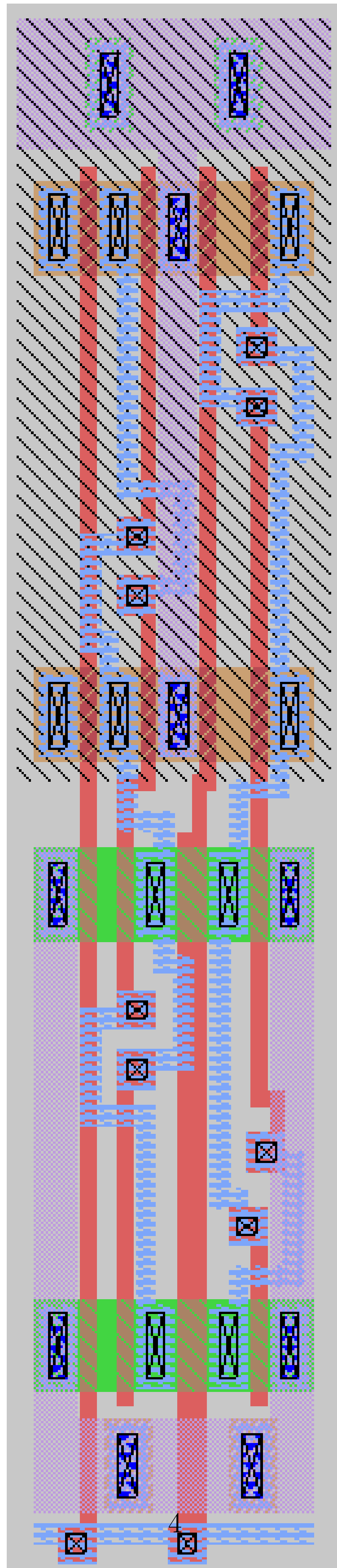

Figure 6: Final layout of the CSRL latch.

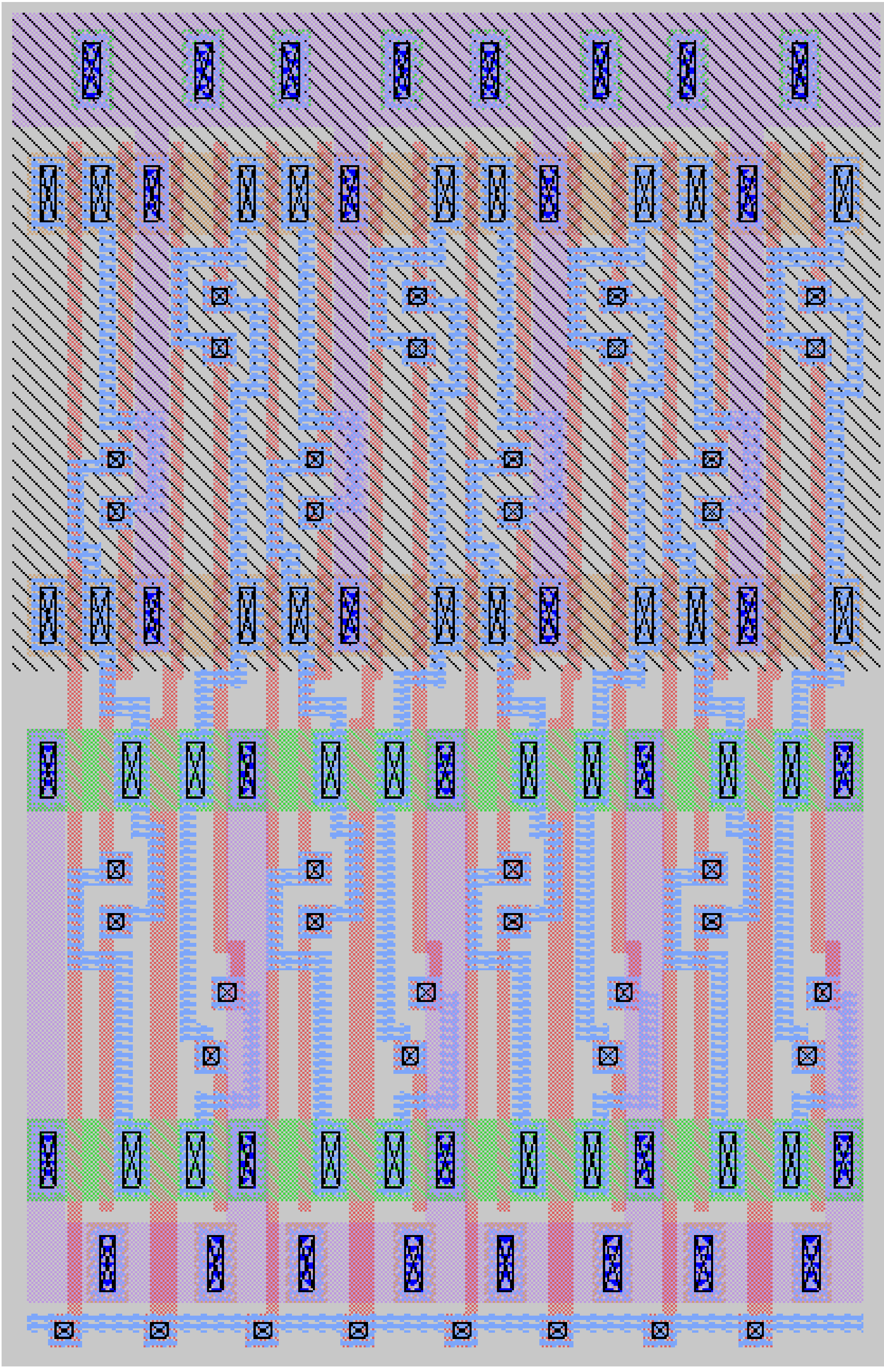

Figure 7: Final layout of the 4 bit shift register. The width of the register is 10.900 microns.

3 Layout Versus Schematic

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Circuit 1 cell sky130_fd_pr__nfet_01v8 and Circuit 2 cell sky130_fd_pr__nfet_01v8 are black boxes.
Equate elements:  no current cell.
Device classes sky130_fd_pr__nfet_01v8 and sky130_fd_pr__nfet_01v8 are equivalent.

Circuit 1 cell sky130_fd_pr__pfet_01v8 and Circuit 2 cell sky130_fd_pr__pfet_01v8 are black boxes.
Equate elements:  no current cell.
Device classes sky130_fd_pr__pfet_01v8 and sky130_fd_pr__pfet_01v8 are equivalent.

Subcircuit summary:
Circuit 1: CSRL_latch                      |Circuit 2: CSRL_Latch
-------------------------------------------|-------------------------------------------
sky130_fd_pr__nfet_01v8 (8)                |sky130_fd_pr__nfet_01v8 (8)
sky130_fd_pr__pfet_01v8 (8)                |sky130_fd_pr__pfet_01v8 (8)
Number of devices: 16                      |Number of devices: 16
Number of nets: 13                         |Number of nets: 13
---------------------------------------------------------------------------------------
Resolving symmetries by property value.
Resolving symmetries by pin name.
Netlists match uniquely.

Subcircuit pins:
Circuit 1: CSRL_latch                      |Circuit 2: CSRL_Latch
-------------------------------------------|-------------------------------------------
D                                          |D
Dn                                         |Dn
VDD                                        |VDD
GND                                        |GND
CLK                                        |CLK
Q                                          |Q
Qn                                         |Qn
---------------------------------------------------------------------------------------
Cell pin lists are equivalent.
Device classes CSRL_latch and CSRL_Latch are equivalent.

Subcircuit summary:
Circuit 1: 4bit_CSRL_latch.spice           |Circuit 2: 4bit_CSRL.spice
-------------------------------------------|-------------------------------------------
CSRL_latch (4)                             |CSRL_Latch (4)
Number of devices: 4                       |Number of devices: 4
Number of nets: 13                         |Number of nets: 13
---------------------------------------------------------------------------------------
Netlists match uniquely.
Cells have no pins;  pin matching not needed.
Device classes 4bit_CSRL_latch.spice and 4bit_CSRL.spice are equivalent.

Final result: Circuits match uniquely.
.
```

Figure 8: Output of netgen checking the netlists of both the schematic and layout. Circuits match uniquely means that are compatible.